



USA TRIATHLON EDUCATION

7-Zone Training Systems

Justin Trolle, USA Triathlon Level 3 Coach

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Overview

A 7-zone system splits training intensity into seven bands anchored to two physiological markers, LT1 and LT2, so every zone boundary lands on something meaningful instead of inside a zone.

Zones are a communication tool, not physiology. The underlying physiology is identical whether a coach uses 3, 5, 6, or 7 zones. What the zones do is make sure the athlete reads a prescription the way the coach meant it.

This 7-zone system keeps LT1 and LT2 as clean borders between zone groups. If you know where LT1, LT2, and VO2 max sit for an athlete, you can build almost any zone model from them.

The seven zones

Zones 1 and 2 sit at or below LT1, Zones 3 and 4 sit between LT1 and LT2, and Zones 5 to 7 sit above LT2.

Zone	Position	Purpose
1	Below LT1	Recovery and easy movement
2	Up to LT1	Aerobic base: capillarization, blood volume, fat use. The classic "Zone 2 work"
3	Between LT1 and LT2	Tempo, moderate aerobic work
4	Just below LT2	Sub-threshold, building toward threshold
5	Just above LT2	Above-threshold work
6	Higher intensity	Anaerobic capacity, lactate tolerance
7	Near VO2 and maximal	Maximal efforts and sprints

Two cautions when you teach this. First, the exact boundary around LT1 varies by source: some place Zone 2 at LT1, others keep Zones 1 and 2 fully below it. Pick one convention and use it consistently across your athletes. Second, many athletes cannot run at true Zone 2 because heart rate spikes even at very slow paces. Two practical fixes help. One is a walk-run protocol or speed walking. The other is to run at the slowest sustainable pace at which running biomechanics are not negatively affected, rather than forcing a pace so slow that form breaks down. For running, the aerobic base often comes more effectively from swimming and cycling.

How 7 zones map to other systems

Every system divides the same lactate curve. The 7-zone model is the most detailed and the easiest to administer because LT1 and LT2 fall between zones.

System	Below LT1	LT1 to LT2	Above LT2	Typical use
3-zone	Zone 1	Zone 2	Zone 3	Swimming, simple prescriptions, beginners
5-zone	Zones 1-2	Zones 3-4	Zone 5	Uncommon in triathlon
6-zone	Zones 1-2 (plus Zone 3 just below LT1)	Zone 4, split by LT2	Zones 5-6	Traditional Olympic-distance model
7-zone	Zones 1-2	Zones 3-4	Zones 5-6-7	Short course, draft-legal, and universal use

The 6-zone model's weakness is that LT2 sits in the middle of Zone 4. An athlete told to ride "Zone 4" could be on either side of threshold with no one noticing, which also muddies comparison of data across coaches and countries.

The extra zones above LT2 became more valuable once sprint and super-sprint team racing arrived around 2014 to 2015.

Applying zones across swim, bike, and run

The zones stay the same in all three sports. Only the name of the LT2 anchor and the way you measure it change.

Sport	LT2 is called	Notes
Swim	Critical Swim Speed (CSS)	Swimming often uses the simpler 3-zone model
Bike	Functional Threshold Power (FTP)	Power or heart rate
Run	Lactate threshold / anaerobic threshold	Heart rate spikes make true Zone 2 hard to hold

These are the same physiological point: the intensity where lactate accumulates faster than the body can clear it, breathing rises noticeably, and exercise duration becomes limited. Lab work calls it OBLA.

Recovery demands also follow the zones. Zones 1 and 2 produce few byproducts and recover quickly. Zones 3 and up need substantially more recovery, and maximal sprint work under 10 seconds needs near-full rest, around 1:5 to 1:8 work to rest, to keep quality.

Work-to-rest ratios by zone

The harder the interval, the longer the rest it needs. Match the rest to the energy system you are targeting, or the session stops producing the adaptation you prescribed.

Zones	Energy system	Recovery demand	Work-to-rest guidance
1-2	Aerobic	Minimal byproducts, quick recovery	Shorter rest. High volume is sustainable with moderate recovery between sessions
3-4	Aerobic into anaerobic	Rising lactate and byproducts	Rest lengthens as intensity approaches LT2. Manage session density and volume
5-6	Anaerobic	Significant lactate and byproducts	Substantially more rest, enough for lactate to partially clear between efforts
7	Alactic / ATP-CP (sprints under 10 seconds)	Rapid depletion, slow replenishment	Near-full recovery, around 1:5 to 1:8 work to rest

Insufficient rest also degrades biomechanics. If technique breaks down during a session, that is a signal the rest periods were too short, and quality work is no longer happening. Many coaches set zones correctly but miss the work-to-rest ratio, and the athlete never gets the intended adaptation despite training in the right zone.

Coaching notes and common mistakes

- **Match the system to the athlete.** Long-course athletes can get by on 5 or 6 zones because they spend little time anaerobically. Short-course and draft-legal athletes need the extra zones above LT2. The 7-zone system covers both.
- **Keep numbering stable.** Zone numbers never change when goals change. Only the zones you emphasize change, and Zones 5 to 7 may simply go unused for a long-course athlete.
- **Scale down for beginners.** Collapse to three zones: easy, moderate, hard.
- **Watch athlete transfers.** An athlete used to a 5-zone system who joins a 7-zone coach can misread prescriptions and train at the wrong intensity. Confirm which system they are on before the first workout.
- **Target limiters.** Athletes slow down because something limits them, not by choice. Prescribe the zones that address the limiter rather than the ones the athlete already does well.
- **Test fresh.** Test in the first week after an adaptation week, then reset zones for the next cycle.